DATA MANAGEMENT CARPENTRY · EPISODE 1

Introduction to Data Management

Why data management matters in research, understanding file systems, and introducing our inherited project scenario.

EPISODE SCOPE

Questions & Objectives

Core Questions

Why is data management important?

How do computers organise files?

What is a file system?

How to navigate with a file explorer?

Learning Objectives

Describe common data management challenges.

Explain file and folder organisation.

Navigate file systems via graphical interfaces.

Locate project data in a file explorer.

CURRICULUM ROADMAP

Working from the Outside In

STEP 1 & 2

Folder Structures

Standard hierarchies & meaningful naming.

STEP 3 & 4

Documentation & Storage

READMEs, workflows, and safe storage.

STEP 5 & 6

Transfer & Selection

Managing transfers; deciding what to keep.

STEP 7 & 8

Tables & Formats

Tidy tabular data & choosing the best file formats.

Focus: Not programming, statistics, or data science — practical skills to find, share, and preserve data.

WORKSHOP NARRATIVE

The Inherited Project Scenario

The Situation

You join a collaborative research project. A researcher recently left the institution, dumping everything into a single folder.

> *"Can you work out what this data is, organise it, and make sure we can continue the project?"*

Throughout this lesson, you take the role of the researcher rescuing this project.

Chaotic Files & Folders

Data!@#$.csv

data_final_v3_LAST_ONE.xlsx

Untitled.csv

New Folder / New Folder (2)

RAW_DATA_!!!

EVERYDAY CHALLENGES

Why Data Management Matters

Small Projects Grow Rapidly

Projects start simply with only a few files, e.g.

experiment.xlsx
notes.docx
results.csv.

As files accumulate, difficulties multiply. Most issues are **organisational**, not technical.

Common Frustrations

"I can't find the file I need."

"Which version is current?"

"Where did this data come from?"

"What does this dataset actually contain?"

COMPLIANCE & GOVERNANCE

Funders, Policies & Legal Duties

Funding Mandates

Detailed Data Management Plans (DMPs) required:

UKRI · Wellcome Trust · NSF · CIHR · Newcastle University institutional policies.

Personal Data & Law

Subject Access: Can you quickly find requested personal data?

Right to Erasure: Can you delete participant data with confidence?

Retention: Keep only what is legally permitted.

DISCUSSION 1

Inherited Files & Old Projects

Have you ever inherited files from another researcher or returned to an old project after several months?

👉 **What was the most difficult part of understanding the data?**

FUNDAMENTAL CONCEPTS

Visualising a File System

Files, Folders & Trees

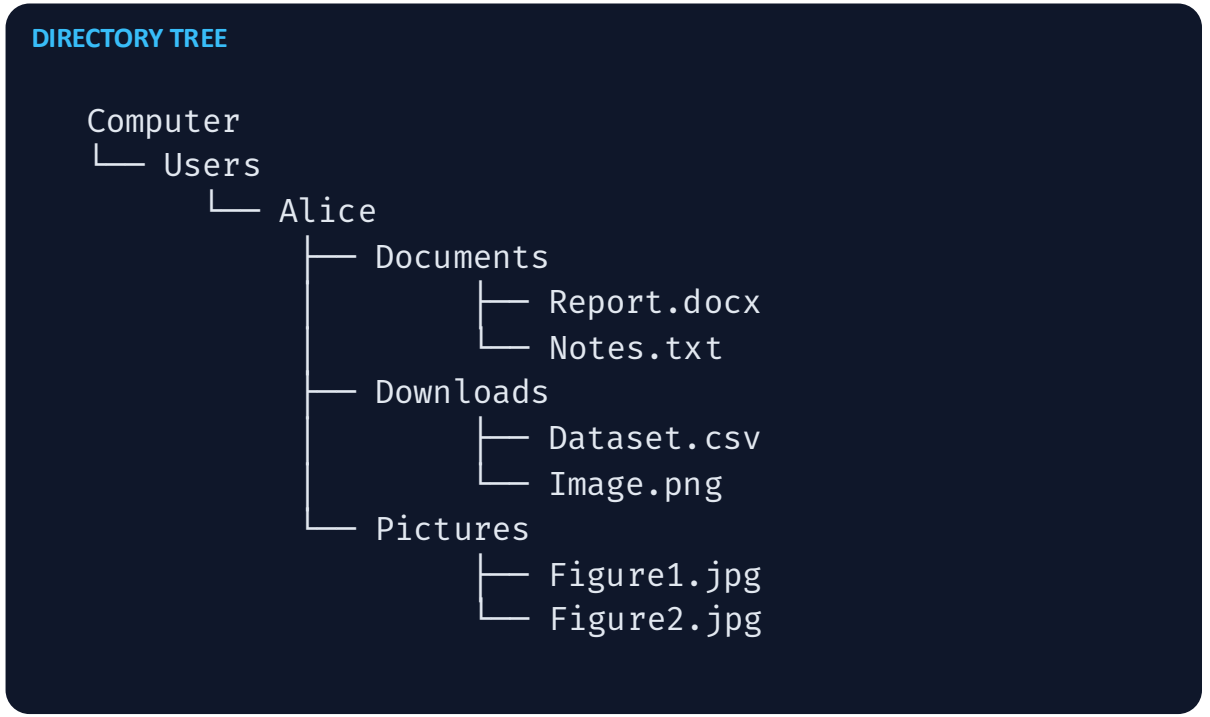
Computers store info in **files**, organised into **folders** (directories), forming a **file system**.

A file system provides

- Structure
- Organisation
- Navigation
- Storage.

Without folders, every file on your computer would exist in one single enormous list.

DIRECTORY TREE



TRANSFERABLE SKILLS

Physical Storage Locations



Internal Drives / SSDs



USB Flash Drives



Network Shares



Cloud Storage

The Key Takeaway:
Regardless of physical storage, files are presented through the same folder-and-file structure. Navigation skills transfer across all systems.

GRAPHICAL TOOLS

Introducing the File Explorer



Windows

File Explorer

Primary tool demonstrated throughout the workshop.



macOS

Finder

Smiling face icon on the dock.



Linux

Files / Nautilus / Dolphin

Varies depending on distribution.

DISCUSSION 2

Opening File Explorer

Windows Options

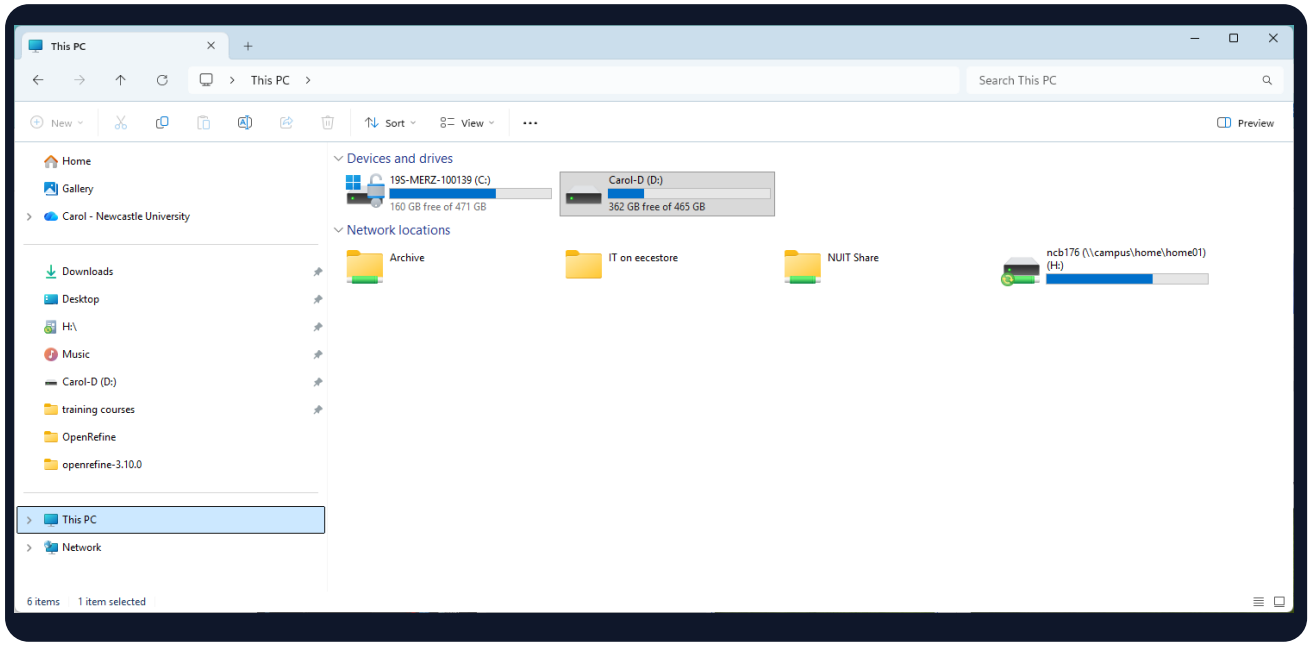
- Click the **Folder icon** on the taskbar
- Use shortcut: Windows Key + E
- Open Start Menu & search **File Explorer**

macOS Options

- Click the **Finder icon** (blue/white face) on dock
- Press Command + Space to search Spotlight

INTERFACE ELEMENTS

Anatomy of File Explorer



Navigation Pane (Left)

Jump quickly between key locations (Desktop, Downloads).

Address Bar (Top)

Shows current folder path and navigation trail.

Search Box (Caution)

Helpful, but never replace good structure with search.

DISCUSSION 3

Navigating the File Explorer

Confirm you can locate on your computer:

KEY COMPONENTS

- Navigation pane
- File listing area
- Address bar
- Search box

REQUIRED ACTIONS

- Move between folders
- Expand and collapse folder trees

EXERCISE

Challenge: Finding the Dataset

Download the lesson dataset and use File Explorer to:

1. Open your **Downloads** folder.
2. Locate the downloaded dataset archive.
3. Open the dataset folder.
4. Explore the contents **without modifying anything**.
5. Identify at least **three things** that seem confusing or poorly organised.

CHALLENGE SOLUTION & REFLECTION

Evaluating the Inherited Dataset

Was It Clear Where Important Data Was?

No. Locating the core findings is hindered by messy naming and loose scripts.

Raw files, generated charts, and draft spreadsheets are indiscriminately lumped together.

What Information Was Missing?

- Documentation explaining dataset contents.
- Clear indication of which data file is authoritative.
- Separation between raw and processed outputs.

SUMMARY

Episode 1 Key Points

Data management is an essential part of research practice.

A well-organised project should be understandable to collaborators and to your future self.

Many common research frustrations arise from poor organisation and documentation.

File explorers provide a graphical way to navigate stored data.

Computers organise information using files and folders within a file system.

Understanding how files are organised is a foundation for good data management.

LOOKING AHEAD

Next: Episode 2

Now that we have located the project files, we will begin improving their folder structure and establishing standard directory hierarchies.

Organising Project Folders